Inequalities

Level 1

Problem 1. Let a, b, c be nonnegative real numbers satisfying $a + b + c = 1$. Prove that

$$a^3 + b^3 + c^3 + 6abc \geqslant \frac{1}{4}.$$

Problem 2. Let a, b, c be nonnegative real numbers. Prove that

$$a^3 + b^3 + c^3 + ab^2 + bc^2 + ca^2 \geqslant 2(a^2b + b^2c + c^2a).$$

Problem 3. Let x, y, z be positive real numbers. Prove that

$$\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y} \geqslant \frac{(x+y+z)^2}{2(xy+yz+zx)}.$$

Problem 4. Let a, b, c be positive real numbers such that $a > c$ and $b > c$. Prove that

$$\sqrt{c(a-c)} + \sqrt{c(b-c)} \leqslant \sqrt{ab}.$$

Problem 5. Let a, b, c, d be positive real numbers. Prove that

$$(a^2 + b^2 + c^2 + d^2)^2 \geqslant (a+b)(b+c)(c+d)(d+a).$$

Problem 6. Let x and y be real numbers. Prove that

$$x^2 + y^2 + 2\sqrt{(x-1)^2 + y^2} \geqslant 1.$$

Problem 7. Let $n \geqslant 2$ and let $a_1, a_2, \ldots, a_n$ be real numbers greater than 1. Prove that

$$(a_1 + 1)(a_2 + 1)\cdots(a_n + 1) < 2^{n-1}(a_1a_2\cdots a_n + 1).$$

Problem 8. Given three distinct nonzero real numbers, prove that we can label them as a, b, c in some order such that

$$\frac{a}{b} + \frac{b}{c} > \frac{a}{c} + \frac{c}{a}.$$

Level 2

Problem 9. Let a, b, c be positive real numbers. Prove the inequality:

$$\frac{a^3}{b^2} + \frac{b^3}{c^2} + \frac{c^3}{a^2} \geqslant \frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a}.$$

Problem 10. Let x and y be nonnegative real numbers. Prove that

$$\frac{1}{(1+x)^2} + \frac{1}{(1+y)^2} \geqslant \frac{1}{1+xy}.$$

Problem 11. Let a, b, c, d be positive real numbers such that $abcd = 1$. Prove that

$$\frac{1+ab}{1+a} + \frac{1+bc}{1+b} + \frac{1+cd}{1+c} + \frac{1+da}{1+d} \geqslant 4.$$

Problem 12. Let x, y, z be nonnegative real numbers. Prove that

$$x^2 + y^2 + z^2 + 2xyz + 1 \geqslant 2(xy + yz + zx).$$

Problem 13. Let $x_1, x_2, \ldots, x_n$ be nonnegative real numbers satisfying

$$x_1 + x_2 + \cdots + x_n \leqslant \frac{1}{2}.$$

Prove that

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geqslant \frac{1}{2}.$$

Problem 14. Find the largest real number k, such that for any positive real numbers a, b,

$$(a+b)(ab+1)(b+1) \geqslant kab^2.$$

Problem 15. Prove that for all positive real numbers a, b, c the following inequality takes place

$$\frac{1}{b(a+b)} + \frac{1}{c(b+c)} + \frac{1}{a(c+a)} \geqslant \frac{27}{2(a+b+c)^2}.$$

Problem 16. Let n be a positive integer and let $a_1, a_2, \ldots, a_n$ be any real numbers. Prove that there exist $m, k \in \{1, 2, \ldots, n\}$ such that

$$\left| \sum_{i=1}^{m} a_i - \sum_{i=m+1}^{n} a_i \right| \leqslant |a_k|.$$